

Chapter 1

Explainability in Reinforcement Learning

Today, the majority of artificial intelligence agents with which humans interact are deployed in controlled or low-risk environments. For example, agents capable of playing games such as Go operate in settings where potential errors remain limited in terms of consequences. Even more autonomous systems, such as agents based on large language models, remain constrained by human supervision mechanisms and are often restricted to tasks such as code generation or information retrieval. Safety-critical applications integrating machine learning components are still rare. Introducing autonomous agents in such environments, requires the establishment of strong guarantees. These requirements include in particular:

- validating the system’s functioning before deployment,
- identifying the origin of a malfunction in case of failure,
- and certifying that an observed failure will not occur again.

The main challenge lies in the fact that the performing agents currently available rely on both reinforcement learning and deep learning. As a result, these agents are unable to provide explanations of the mechanisms underlying their decisions. This lack of explainability makes their use difficult in safety-critical contexts. To address this limitation, a research field has emerged: Explainable Reinforcement Learning (XRL), a subfield of Explainable Artificial Intelligence (XAI).

This chapter aims to present the main existing XRL techniques. To this end, we first define what is meant by an “explanation” in the context of XAI. We then propose a taxonomy for organizing the XRL research field. Finally, following this organization, we review the state of the art.

1.1 Explanation in Artificial Intelligence

In this section, we define what an explanation is in the context of XAI and how it can be obtained. We first introduce the concept of explanation in a general. We then describe how this concept applies to XAI. Finally, we introduce the

31 concept of explainability methods and describe the properties that characterize
 32 explanations in the context of XAI.

33 1.1.1 Explanation

34 Establishing what is meant by an explanation is challenging due to the funda-
 35 mentally multidisciplinary nature of the concept. When we refer to explanation,
 36 we are inherently referring to an explanation provided to a human. As soon as
 37 humans are involved, the problem is no longer purely computational or math-
 38 ematical. It involves several fields of research, such as philosophy, psychology,
 39 and sociology. So far, these disciplines have not converged on a unified definition
 40 of explanation that can be directly applied to XAI.

41 For this reason, we propose in this manuscript a definition of the concept of
 42 explanation. According to this definition, an explanation involves two entities:
 43 the object (Definition 1.1.1) being explained and the receiver of the explanation.
 44 The receiver is a human who aims to understand how the object operates and will
 45 be referred to as the observer throughout this manuscript (Definition 1.1.2). The
 46 observer seeks to understand the object and, in doing so, builds a mental model
 47 of its functioning. An explanation is therefore what allows the observer to refine
 48 and correct their mental model of how the object operates (Definition 1.1.4).

49 For an observer to understand an explanation, the information it contains
 50 must be interpretable (Definition 1.1.3). Information is considered interpretable
 51 when it can be directly understood by a human without requiring any additional
 52 transformation process.

Definition 1.1.1: Object

The object of an explanation is the system is to be understood by an observer.

Definition 1.1.2: Observer

The observer is the human who aims to understand a object. The observer
 possesses a mental model of the object being explained. This mental model
 is an internal representation of how the observer believes the system operates.
 This representation may be incomplete or inaccurate.

Definition 1.1.3: Interpretable

Information is interpretable when it can be directly understood by an
 observer without requiring any additional transformation process.

Definition 1.1.4: Explanation

An explanation is any interpretable information that enables an observer to
 refine or correct their mental model of the object of the explanation.

56 This definition of explanation provides the foundation for the approach
 57 to explainability adopted throughout this manuscript. Its generality allow it
 58 to encompass the different explainability methods considered in the following
 59 sections. Moreover, it avoids a binary interpretation of explanation, where a
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system is considered either explainable or not explainable. From our perspective, the notion of non-explainable system is not meaningful.

Considering a system as impossible to explain would therefore contradict the general objective of scientific understanding. This quantitative perspective allows us to express an intuitive idea: not all explanations provide the same level of value. The effectiveness of an explanation depends on its ability to improve the observer's understanding of the system. For example, describing a marginal aspect of a system is different from correcting an observer's misunderstanding of a key mechanism.

Now that the general framework of explanation has been established, we can examine how this concept applies in the context of artificial intelligence.

1.1.2 Explainable Artificial Intelligence

In the context of XAI, the object of explanation is an artificial intelligence system. More precisely, XAI emerged as a response to the success of deep learning models. These models achieve high performance through connectionist architectures based on deep neural networks. In these architectures, information is represented in a distributed manner across the network through patterns of activation and connections between computational units. Unlike symbolic approaches, individual components of neural networks cannot be directly associated with human-defined concepts. The knowledge encoded by these models cannot be understood simply by inspecting their structure.

From this starting point, two types of approaches emerge:

- **Substitutive explainability:** In this approach, the underlying assumption is that there is nothing to explain within a neural network. There is no semantic meaning associated with each computational component, and therefore nothing to interpret. This corresponds to the so-called "black box" view of neural networks. Under this assumption, XAI consists of replacing as much of the deep learning system as possible with alternative AI methods.
- **Intrinsic explainability:** In this approach, it is argued that neural networks inherently contain mechanisms that can be explained. The limitation is considered to lie not in the networks themselves, but in the absence of adequate tools to analyze their functioning.

In this manuscript, we present methods stemming from both approaches to explainability. Both are indeed capable of generating explanations for AI systems. However, to be transparent with the reader, this manuscript follows the intrinsic explainability perspective.

1.1.3 Explainability Method

Now that the general framework of XAI has been established, we need to further detail the properties of an explanation in this context. In our framework, an explanation in the field of XAI can be characterized by three components:

- **Source:** refers to the origin of the information used by the explainability method. The explainability method transforms the raw information pro-

duced by the AI system into an explanation that can be understood by an observer.

- **Form:** refers to the representation through which the explanation is communicated. The form defines how the information is presented to the observer and must allow human interpretation. An explanation can take different forms, such as text, images, computer code, mathematical equations.
- **Subject:** refers to the aspect of the AI system that is being explained. In XAI, this distinction is commonly associated with the difference between local and global explanations. A local explanation describes why an AI system produced a specific output in a given situation. A global explanation aims to describe the general functioning of the AI system.

An explainability method can therefore be viewed as a transformation process that maps a source of information into an explanation characterized by a specific form and subject. Current research in XAI focuses on developing new methods capable of performing this transformation in order to overcome the lack of interpretability associated with deep learning models.

1.2 Taxonomy of XRL Methods

Now that the concept of XAI has been defined, we turn to its application in reinforcement learning, known as XRL. As in XAI, explanations in XRL are produced by explainability methods and can be described in terms of their subject, source, and form. This section reviews the possible variations of these three aspects in the specific context of reinforcement learning.

1.2.1 Subject

The first question to address is the subject of explanation in the context of reinforcement learning. To answer this question, it is necessary to identify the different components involved in a reinforcement learning system and determine which of them constitutes the object of explanation. A reinforcement learning system can be described through two main components: the environment and the policy.

The environment defines the dynamics in which the agent evolves, including the possible states, observations, actions, and transitions. Environments are simulations designed by humans, and their dynamics are explicitly defined. Therefore, it is assumed that the environment is known to the observer and does not require explanation.

The main object of interest is therefore the agent's policy, denoted π . The policy represents the decision-making mechanism of the agent. It is a function that maps observations $\mathcal{O}$ to actions $\mathcal{A}$ and is learned through interaction with the environment in order to maximize the expected cumulative reward. The policy is the component that encodes the learning process and that XRL aims to explain.

Explaining a policy introduces additional challenges compared with classical XAI settings. In traditional XAI problems, the objective is to explain an isolated

147 decision, such as determining which features of an image led to a classification
 148 result. In contrast, reinforcement learning introduces a temporal dimension
 149 into the decision-making process. An action is not only determined by the
 150 current observation but also influences future states and rewards. Therefore,
 151 understanding the behavior of an agent may require considering a sequence of
 152 interactions between the agent and its environment.

153 This temporal dimension leads to a distinction between two objectives of
 154 explanation in XRL:

- 155 • **Decision explanations:** focus on identifying the elements of the current
 156 observation that influenced the action selected by the policy. This type of
 157 explanation is close to classical XAI approaches, as it analyzes the rela-
 158 tionship between inputs and outputs while abstracting away the sequential
 159 nature of the decision process.
- 160 • **Strategy explanations:** aim to characterize the agent’s behavior over
 161 time. They focus on sequences of decisions and interactions with the
 162 environment in order to understand how the agent achieves its long-term
 163 objective. This type of explanation is specific to reinforcement learning, as
 164 it addresses the temporal and goal-oriented nature of the learning process.

165 1.2.2 Source

166 We identify three sources of explanation in reinforcement learning, all obtained
 167 during or after the learning process and containing information about the agent:

- 168 • **Policies:** policies are functions that generate a distribution over the action
 169 space for a given observation. The objective of reinforcement learning is
 170 to learn a policy that maximizes the expected cumulative future rewards.
 171 During the learning phase, the policy is iteratively improved through
 172 interactions with the environment.
- 173 • **Trajectories:** trajectories represent sequences of successive interactions
 174 between the agent and the environment. At each step, the agent selects
 175 an action based on its observation, which modifies the environment and
 176 produces a reward as well as a new observation. This process continues
 177 until the end of the trajectory.
- 178 • **Value functions:** value functions provide predictions about the future
 179 outcome of a trajectory from a given observation. The most commonly
 180 used value function is the critic function, which estimates the expected
 181 sum of future rewards. This prediction guides the learning process, as the
 182 agent updates its policy to maximize the estimated future return. However,
 183 other types of value functions can also be used as sources of explainability.

184 1.2.3 Form

185 We identify six forms of explanation in the RL setting:

- 186 • **Policy Model:** an interpretable model of the agent’s policy function.
- 187 • **Saliency Map:** a weighting of the input features representing their
 188 influence on the output of the policy function.

- 189 • **Counterfactual:** a representation of a modified observation obtained by
190 perturbing the original observation in order to produce a different action
191 from the agent.
- 192 • **Agent Trajectory:** a representation of a sequence of successive interac-
193 tions between the agent and the environment.
- 194 • **Trajectory Prediction:** a prediction about the future evolution of the
195 trajectory produced by a value function.
- 196 • **Interaction Model:** an interpretable model of the interactions between
197 the agent and the environment.

198 Among the three components characterizing an explanation, the form is the
199 one that varies most significantly across explainability methods. While subjects
200 and sources tend to be relatively stable in the XRL literature, the diversity in
201 how explanations are presented is considerably greater. The list above is by no
202 means definitive and may evolve as new approaches emerge. For this reason, the
203 following section is organized according to the form of explanations and presents
204 existing explainability methods based on this criterion.